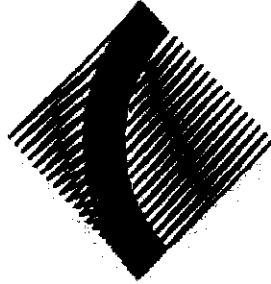


AW  
AT  
JG  
LB  
KA  
KW  
KP  
FH

Name: \_\_\_\_\_  
Class: 12MT2\_\_ or 12MTX\_\_  
Teacher: \_\_\_\_\_

## CHERRYBROOK TECHNOLOGY HIGH SCHOOL



2009 AP4

YEAR 12 TRIAL HSC EXAMINATION

# MATHEMATICS

*Time allowed - 3 HOURS  
(Plus 5 minutes reading time)*

### DIRECTIONS TO CANDIDATES:

- Attempt all questions.
- All questions are of equal value.
- Each question is to be commenced on a new page clearly marked Question 1, Question 2, etc on the top of the page. \*\*.
- All necessary working should be shown in every question. Full marks may not be awarded for careless or badly arranged work.
- Approved calculators may be used. Standard Integral Tables are provided
- Your solutions will be collected in one bundle stapled in the top left corner.

Please arrange them in order, Q1 to 10.

***\*\*Each page must show your name and your class. \*\****

**Question 1 (12 marks)****Marks**

- (a) Expand and simplify  $8x^2 + 2xy - 2x(4x + y)$ . **2**
- (b) Completely factorise  $5x^2 - 20$  **2**
- (c) Find the value of  $\tan \frac{8\pi}{7}$  correct to three significant figures. **2**
- (d) Find the values of  $x$  for which  $|2 - 3x| > 11$ . **2**
- (e) Express  $3\sqrt{32} - \sqrt{128}$  in simplest surd form. **2**
- (f) If  $k = \frac{1}{3}m(v^2 - u^2)$  find the value of  $m$  when  $k = 1224$ ,  $v = 14.2$  and  $u = 7.4$ . **2**

**Question 2** (12 marks) ( START A NEW PAGE )

**Marks**

(a) Differentiate with respect to  $x$ .

(i)  $(2x^2 + 1)^8$

2

(ii)  $x^2 e^{2x}$

2

(iii)  $\frac{x}{\ln x}$

2

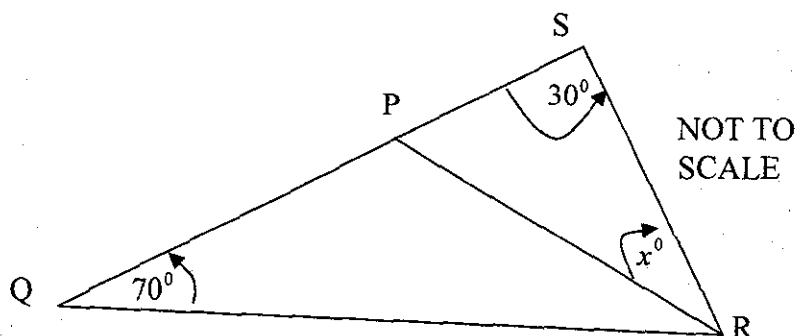
(b) (i) Find  $\int (2x+1)^5 dx$

2

(ii) Evaluate  $\int_0^{\frac{\pi}{8}} 4\cos 4x dx$

2

(c)



Given  $PQ = PR$ ,  $\angle PQR = 70^\circ$ ,  $\angle PSR = 30^\circ$  and  $\angle PRS = x^\circ$ ,

2

find the value of  $x$  giving reasons.

**Question 3** (12 marks)      ( START A NEW PAGE )

**Marks**

(a) Given the points A(1,2), B(3,1), C(-1,4)

(i) Find the equation of the line BC.

1

(ii) Find the perpendicular distance from point A to the line BC.

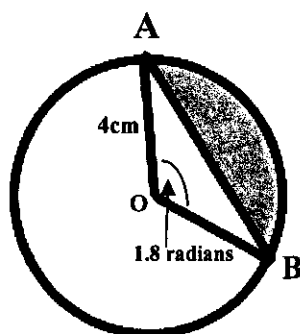
2

(iii) Hence, or otherwise, find the area of  $\triangle ABC$ .

2

(b) A chord AB subtends an angle of  $1.8$  radians at the centre of a circle of radius  $4$  cm. Find the area of the minor segment cut off by the chord AB. ( Give your answer correct to 1 decimal place.)

2



(c) For the function  $y = 2\sqrt{9 - x^2}$

state the

(i) domain      and

2

(ii) range.

1

(d) Find the radius and centre of the circle with equation

2

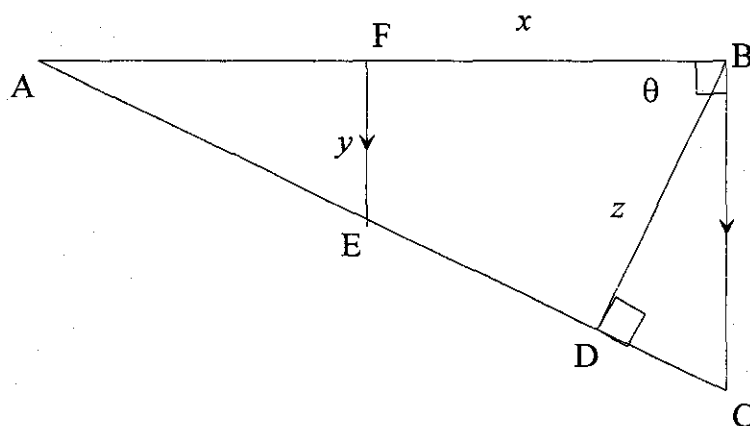
$$x^2 - 4x + y^2 + 6y + 12 = 0.$$

**Question 4 (12 Marks)**

( START A NEW PAGE )

**Marks**

- (a) The right triangle ABC is shown below.  $BC \parallel FE$ ,  $BD \perp AC$ ,  $\angle FBD = \theta$ ,  $BF = x$ ,  $EF = y$  and  $BD = z$ .



Prove

(i)  $\triangle BDA \parallel \triangle EFA$

2

Show that

(ii)  $\angle FEA = \theta$

1

(iii)  $AB = x + y \tan \theta$

1

(iv)  $z = x \cos \theta + y \sin \theta$

1

- (b) Find the equation of the normal to the curve  $y = \sqrt{x}$  at the point (4,2).

2

- (c) A ship sails from port A, 60 nautical miles due west, to a port B. It then proceeds a distance of 50 nautical miles on a bearing of  $210^\circ$  to a port C.

- (i) Draw a diagram to illustrate the information given.

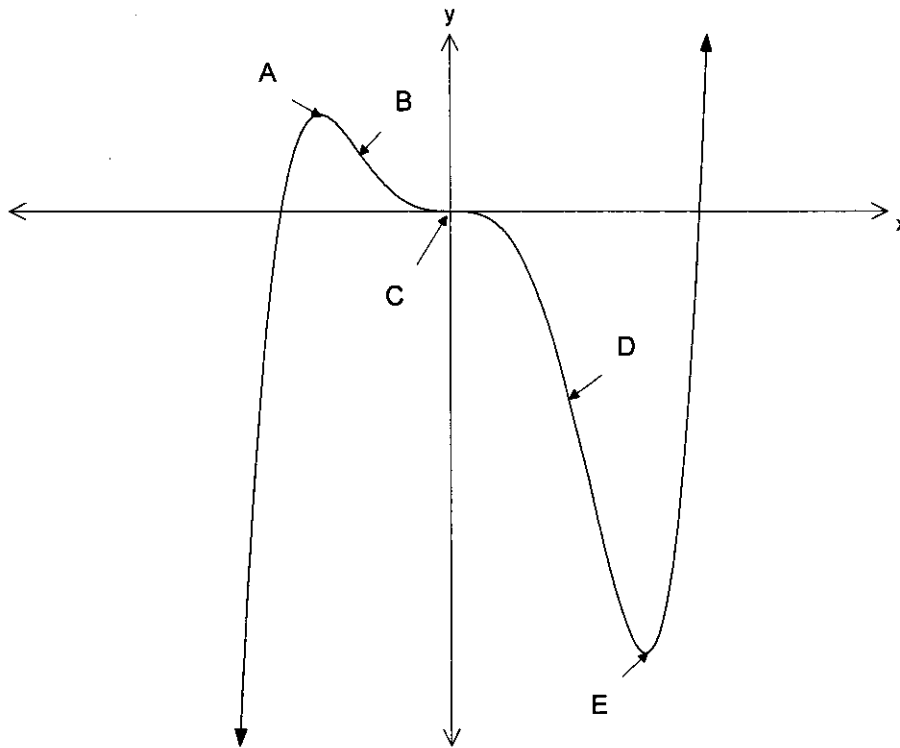
1

- (ii) Find the distance (nearest nautical mile) and bearing of C from A.

4

**Question 5 (12 Marks)****( START A NEW PAGE )****Marks**

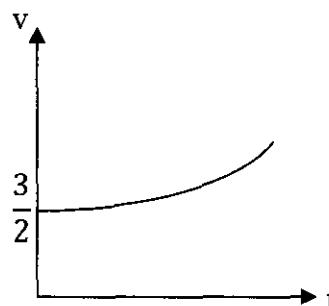
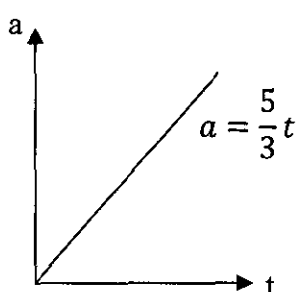
- (a) The graph of the curve  $y = f(x)$  is drawn below.



- |       |                               |   |
|-------|-------------------------------|---|
| (i)   | Name the points of inflexion. | 1 |
| (ii)  | Name the stationary points.   | 1 |
| (iii) | Sketch the gradient function. | 1 |
- (b) The rate at which Carbon Dioxide will be produced when conducting an experiment is given by  $\frac{dV}{dt} = \frac{1}{100}(30t - t^2)$  where  $V \text{ cm}^3$  is the volume of gas produced after  $t$  minutes.
- |      |                                                                                |   |
|------|--------------------------------------------------------------------------------|---|
| (i)  | At what rate is the gas being produced 15 minutes after the experiment begins? | 1 |
| (ii) | How much Carbon Dioxide has been produced during these 15 minutes?             | 2 |

Question 5 is continued on page 6.....

- (c) The graphs below describe the flight of a sugar glider after it jumps from a tall tree. Assume that the displacement at time  $t = 0$  is 0 metres.



- (i) Show that the velocity of the sugar glider for  $t \geq 0$  is given by  $v = \frac{1}{6}(5t^2 + 9)$ . 2
- (ii) Determine the expression for the displacement ( $x$ ) for  $t \geq 0$  and hence calculate the displacement of the sugar glider at  $t = 2$ . 2
- (d) The student population “ $C$ ” at Cherrybrook Technology High School in the first 6 months of 2009 was increasing at a decreasing rate. What does this tell you about

$$\frac{dC}{dt} \text{ and } \frac{d^2C}{dt^2} \quad . \quad \quad \quad 2$$

**Question 6 (12 Marks)****( START A NEW PAGE )****Marks**

- (a) For the curve  $y = 2x^3 - 9x^2 + 12x$ ,
- (i) Show that  $\frac{dy}{dx} = 6(x-1)(x-2)$  1
  - (ii) Find the coordinates of the stationary points and determine their nature. 3
  - (iii) Graph the function showing clearly the stationary points. 2
  - (iv) Where is  $y=f(x)$  decreasing? 1
- (b) Consider the parabola  $y^2 = 8(x+2)$
- (i) Find the coordinates of the vertex. 1
  - (ii) Find the coordinates of the focus. 1
  - (iii) Find the equation of the directrix. 1
- (c) Find the value(s) of  $m$  for which the equation  $4x^2 - mx + 9 = 0$  has exactly one real root. 2

- (a) The population,  $P$ , of a certain town grows at a rate proportional to the population, ie.  $\frac{dP}{dt} = kP$ .

If the population grows from 20 000 to 25 000 in two years, find:

- (i) the value of  $k$  (the growth constant) correct to 4 significant figures. 2
- (ii) the population of the town, to the nearest hundred, after a further 8 years. 1
- (iii) Calculate the rate of change of the population at this time (ie after these further 8 years) correct to the nearest hundred. 1

- (b) A particle moves in a straight line so that its displacement,  $x$  metres from a fixed point  $O$  on a line, is given by

$$x = t + \frac{16}{t+1}$$

where  $t$  is measured in seconds.

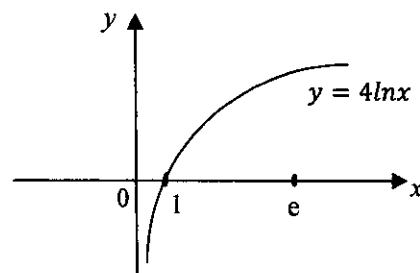
- (i) Find the particle's initial position 1
- (ii) Find expressions for the velocity and acceleration of the particle in terms of  $t$ . 2
- (iii) Find when and where the particle is at rest. 2
- (iv) Find the limiting velocity of the particle. 1
- (v) Find the total distance travelled by the particle in the first 4 seconds. 2

**Question 8 (12 Marks)****( START A NEW PAGE )****Marks**

- (a) The gradient of a curve is given by  $\frac{dy}{dx} = \frac{2x}{x^2 + e}$  . 2  
The curve passes through the point (0,2).

What is the equation of the curve?

- (b) The arc of the curve  $y = 4 \ln x$ , between  $x = 1$  and  $x = e$ , is rotated about the y-axis.



- (i) Show that the volume formed is given by  $V = \pi \int_0^4 e^{0.5y} dy$  2
- (ii) Hence, find the exact volume. 2
- (c) (i) Is the series  $\log 3 + \log 9 + \log 27 + \dots$  arithmetic or geometric? 2  
Give reasons for your answer.
- (ii) Find the sum of the first 10 terms of the series in exact form. 2
- (d) Use the trapezoidal rule with four function values. to find an approximate value of 2  
the area under the curve  $y = 3^x$ , bounded by the x axis,  $x = 1$  and  $x = 4$

**Question 9** (12 marks) ( START A NEW PAGE )

**Marks**

(a) Solve  $2\cos x + \sqrt{3} = 0$  for  $-\pi \leq x \leq \pi$  2

(b) (i) Differentiate  $e^{\tan x}$  1

(ii) Hence or otherwise, evaluate  $\int_0^{\frac{\pi}{4}} 3e^{\tan x} \sec^2 x \, dx$  2

(c) The minute hand on a clock face is 12 centimetres long.

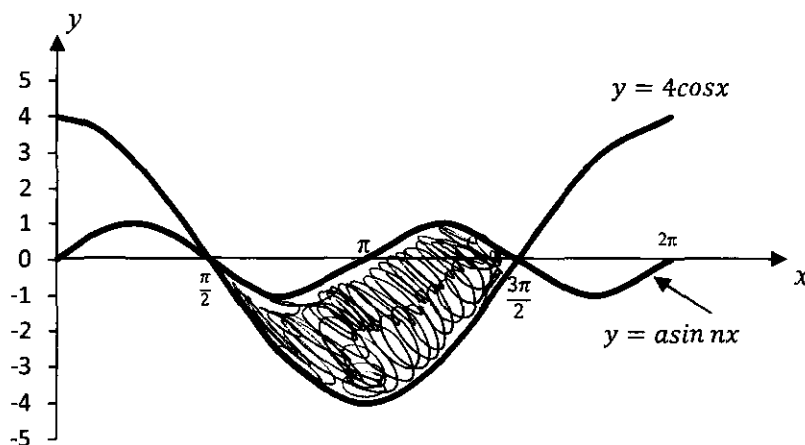
In 40 minutes,

(i) through what angle does the hand move (in radians)? 1

(ii) how far does the tip of the hand move? 1

(iii) what area does the hand sweep through in this time? 1

(d)



The graph above shows  $y = 4\cos x$  and  $y = a\sin nx$ .

(i) Find the values of  $a$  and  $n$ . 1

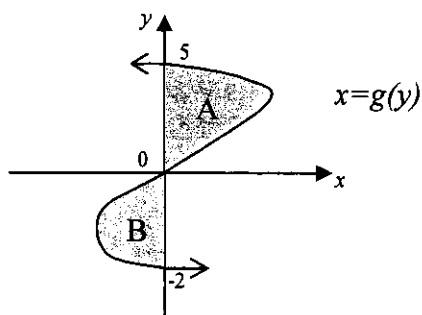
(ii) Find the area of the shaded region in the diagram. 3

**Question 10** (12 marks) ( START A NEW PAGE )

**Marks**

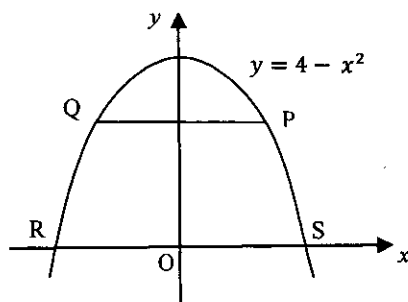
- (a) The area A is equal to 7 square units and area B is equal to 9 square units.

**1**



Evaluate  $\int_{-2}^5 g(y) dy$ .

- (b)



Not to scale

The parabola  $y = 4 - x^2$  cuts the  $x$ -axis at R and S. The point  $P(x,y)$  lies on the parabola in the first quadrant. Q also lies on the parabola such that PQ is parallel to the  $x$ -axis.

- (i) Write down the co-ordinates of R and S.

**1**

- (ii) Show that the area of trapezium PQRS is given by:

**2**

$$A = (2 + x)(4 - x^2)$$

- (iii) Hence, find the value of  $x$  which gives a maximum value of  $A$ , justifying your answer.

**3**

Question 10 is continued on page 12.....

- (c) A small oil company has 50 000 barrels of oil in stock. In its first month of operation, the company increased its stock by 5%, and then sold 1500 barrels.

Each subsequent month, the company continued to increase its previous month's stock by 5%, and then also increased the number of barrels of oil sold by 3%.

- (i) Show that the number of barrels of oil in stock after 3 months can be expressed as:

3

$$B_3 = 50000 \times 1.05^3 - 1500 \times (1.05^2 + 1.05 \times 1.03 + 1.03^2)$$

- (ii) Show that the number of barrels of oil in stock after  $n$  months can be expressed as:

2

$$B_n = 50000 \times 1.05^n - 75000 \times 1.05^n \left( 1 - \left( \frac{1.03}{1.05} \right)^n \right)$$

End of Examination.

## STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

NOTE :  $\ln x = \log_e x, x > 0$

SOLUTIONSQuestion 1

$$\begin{aligned}
 (a) \quad & 8x^2 + 2xy - 2x(4x + y) \\
 & = 8x^2 + 2xy - 8x^2 - 2xy \\
 & = 0
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad & 5x^2 - 20 = 5(x^2 - 4) \\
 & = 5(x - 2)(x + 2)
 \end{aligned}$$

$$\begin{aligned}
 (c) \quad & \tan \frac{8\pi}{7} = 0.4815746, \dots \\
 & = 0.482
 \end{aligned}$$

$$(d) \quad |2 - 3x| > 11$$

$$2 - 3x < -11 \quad \text{or} \quad 2 - 3x > 11$$

$$-3x < -13 \quad \text{or} \quad -3x > 9$$

$$x > \frac{13}{3} \quad \text{or} \quad x < -3$$

$$\begin{aligned}
 (e) \quad & 3\sqrt{32} - \sqrt{128} = 12\sqrt{2} - 8\sqrt{2} \\
 & = 4\sqrt{2}
 \end{aligned}$$

$$(f) \quad K = \frac{1}{2}m(v^2 - u^2)$$

$$1224 = \frac{1}{2}m(14.2^2 - 7.4^2)$$

$$1224 = 48.96m$$

$$m = \frac{1224}{48.96}$$

$$m = 25$$

Question 2

(a) (i) =  $8(2x^2+1)^7 \times 4x$  ✓

=  $32x(2x^2+1)^7$  ✓

award (2)  
for answer only

(ii)  $x^2 \times 2e^{2x} + e^{2x} \times 2x$  ✓

=  $2x^2e^{2x} + 2xe^{2x}$   
or =  $2xe^{2x}(x+1)$  ✓

award (2)  
answer only

(iii)  $\ln x \times 1 - x \times \frac{1}{x}$

$\frac{(\ln x)^2}{(\ln x)^2}$  ✓

do not accept  $\ln x^2$   
(if correct with  $\ln x^2$  give (1))

award (2)  
answer only

(b) (i)  $\int (2x+1)^5 dx = \frac{(2x+1)^6}{6 \times 2} + c$  ✓

ignore missing +c

=  $\frac{1}{12}(2x+1)^6 + c$  ✓

award (2)  
answer only

(ii)  $\int_0^{\pi/8} 4\cos 4x dx$   
=  $\left[ \sin 4x \right]_0^{\pi/8}$  ✓  
=  $\sin \frac{\pi}{2} - \sin 0$   
= 1 ✓

(c)  $PQ = QR$

$\therefore \angle PQR = \angle PRQ = 70^\circ$  (In any  $\Delta$ , angles opposite = sides are) ✓

$\Delta PQR$  IS ISOSCELES ( $PQ = QR$ )  
 $\therefore \angle PQR = \angle PRQ = 70^\circ$  (equal base angles of isosceles  $\Delta$ )

In  $\Delta QSR$

$70 + 70 + x + 30 = 180$   
(angle sum  $\Delta$ ) ✓

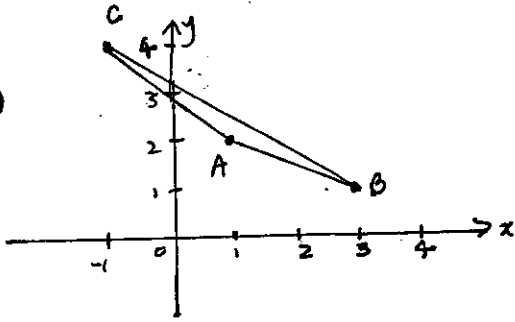
$\therefore x = 10$

for 2nd mark  
needs to see some more + words angle sum  $\Delta$  then  $x=10$

for 1st mark  
must all do this way  
need to see  
isos because of = sides  
 $\angle PRQ = 70^\circ$  (equal base angles)

### QUESTION 3

(a) (i)



$$\text{slope } BC = -\frac{3}{4}$$

equation BC

$$y - y_1 = m(x - x_1)$$

$$y - 4 = -\frac{3}{4}(x + 1)$$

$$4y - 16 = -3x - 3$$

$$3x + 4y - 13 = 0 \quad \checkmark \quad (\text{or equivalent expression})$$

$$(ii) \quad p = \frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}}$$

$$= \frac{|3 \times 1 + 4 \times 2 - 13|}{\sqrt{3^2 + 4^2}}$$

$$= \frac{|3 + 8 - 13|}{\sqrt{25}}$$

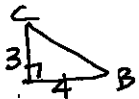
$$= \left| \frac{-2}{5} \right|$$

$$= \frac{2}{5} \text{ units}$$

$x_1, y_1$   
 $A(1, 2)$

correct  
but no  
absolute  
sign  
Ans ①

(iii) Find  $\underline{BC}$  using Pythagoras' Theorem or distance Formula



$$BC = 5 \text{ units}$$

(Pythagorean Triad {3, 4, 5})

$$\text{Area} = \frac{1}{2} \times \frac{3}{5} \times 5$$

$$= 1 \text{ sq. unit}$$

$$(b) \text{ Area} = \frac{1}{2} r^2 (\theta - \sin \theta)$$

$$= \frac{1}{2} \times 4^2 \times (1.8 - \sin 1.8)$$

$$= 6.60921 \dots$$

$$= 6.6 \text{ cm}^2 \text{ (1 dec pl)}$$

$$(c) (i) 9 - x^2 \geq 0$$

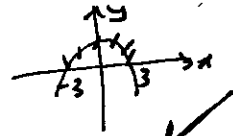
$$(3 - x)(3 + x) \geq 0$$

$$-3 \leq x \leq 3$$

(or  $x \geq -3$  and  $x \leq 3$ )

$$(ii) 0 \leq y \leq 6$$

(or  $y \geq 0$  and  $y \leq 6$ )



$$(d) x^2 - 4x + y^2 + 6y + 12 = 0$$

$$x^2 - 4x + 4 + y^2 + 6y + 9 = -12 + 4 + 9$$

$$(x - 2)^2 + (y + 3)^2 = 1$$

$\therefore \text{radius} = 1 \text{ unit}$   
 $\text{centre} = (2, -3)$

# QUESTION 4

(a) (i) In  $\triangle BDA$ ,  $EFA$   
 $\angle BDA = 90^\circ$  (given  $BD \perp AC$ )  
 $\angle AFE = \angle ABC$  (corresponding angles,  $FE \parallel BC$ )  
 $= 90^\circ$

- $\angle AFE = \angle BDA = 90^\circ$
- $\angle BAD$  is common
- $\therefore \triangle BDA \parallel \triangle EFA$  (equiangular)

(ii)  $\angle DBA = \angle FEA = \theta$   
 (corresponding angles of similar  $\triangle$ 's)

(iii) In  $\triangle AFE$   
 $\tan \theta = \frac{AF}{y}$   
 $AF = y \tan \theta$   
 $\therefore AB = y \tan \theta + x$

(iv) In  $\triangle ABD$   
 $\cos \theta = \frac{z}{AB}$   
 $= \frac{z}{y \tan \theta + x}$

$$z = y \tan \theta \cos \theta + x \cos \theta$$

$$= y \frac{\sin \theta}{\cos \theta} \cos \theta + x \cos \theta$$

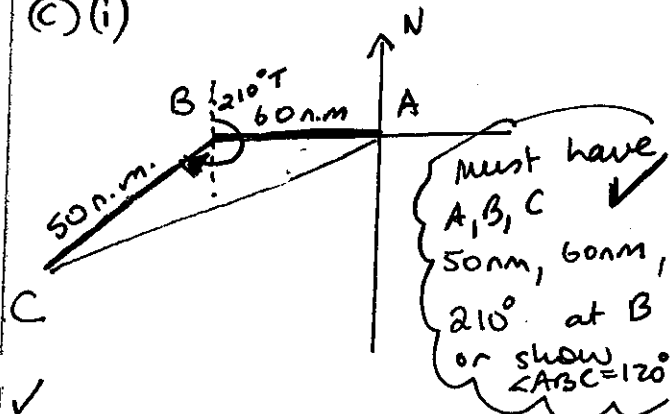
$$= y \sin \theta + x \cos \theta$$

(b)  $\frac{dy}{dx} = \frac{1}{2} x^{-\frac{1}{2}}$   
 $= \frac{1}{2\sqrt{x}}$

when  $x = 4$   
 slope of tangent  $= \frac{1}{2\sqrt{4}} = \frac{1}{4}$   
 $\therefore$  slope of normal  $= -4$  ( $m_1 m_2 = -1$ )

$x = 4, y = 2$   
 $y - 2 = -4(x - 4)$   
 $y - 2 = -4x + 16$   
 $4x + y - 18 = 0$   
 or  
 $y = -4x + 18$   
 or  
 (equivalent eqn)

(c) (i)



(d)  $AC^2 = 60^2 + 50^2 - 2 \times 60 \times 50 \times \cos 120^\circ$

$AC = 95.3939 \dots$   
 $AC = 95 \text{ n.m. (nearest nm)}$

$\frac{\sin \angle BAC}{50} = \frac{\sin 120^\circ}{AC}$

$\sin \angle BAC = \frac{50 \sin 120^\circ}{AC}$

$\angle BAC = 27^\circ$

Bearing of C from A is  $243^\circ \text{T}$  or  $563^\circ \text{W}$

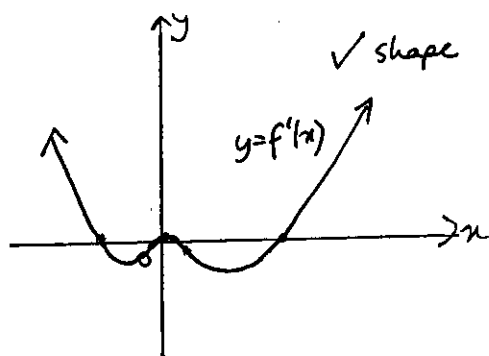
(answer using 95nm rounded is same  $\rightarrow$  Accept!)

# QUESTION 5

(a) (i) B, C and D ✓

(ii) A, C and E. ✓

(iii)



$$(b) \frac{dv}{dt} = \frac{1}{100} (30t - t^2)$$

$$(i) \frac{dv}{dt} = \frac{1}{100} (30 \times 15 - 15^2)$$

$$= 2\frac{1}{4}$$

$$2\frac{1}{4} \text{ cm}^3/\text{min}$$

$$(ii) \int_0^{15} \frac{1}{100} (30t - t^2) dt$$

$$= \frac{1}{100} \left[ 15t^2 - \frac{t^3}{3} \right]_0^{15}$$

$$= \frac{1}{100} \left[ (15 \times 15^2 - \frac{15^3}{3}) - (0) \right]$$

$$= \frac{1}{100} \left[ 3375 - \frac{3375}{3} \right]$$

$$= 22.5 \checkmark$$

$\therefore 22.5 \text{ cm}^3$  is produced in 15 minutes

OR  $v = \int \frac{1}{100} (30t - t^2) dt$

$$= \frac{1}{100} \left[ 15t^2 - \frac{t^3}{3} \right] + c \quad (\checkmark)$$

$$t=0 \quad v=0$$

$$0 = \frac{1}{100} [0 - 0] + c$$

$$c=0$$

$$\therefore v = \frac{1}{100} \left[ 15t^2 - \frac{t^3}{3} \right]$$

$$t=15 \quad v = \frac{1}{100} \left[ 15 \times 15^2 - \frac{15^3}{3} \right]$$

$$= 22.5 \text{ cm}^3 \quad (\checkmark)$$

alternate method and corresponding mark allocation

$$(c) (i) a = \frac{5}{3}t$$

$$v = \frac{5}{6}t^2 + c$$

$$t=0, v = \frac{3}{2} \text{ (diagram)}$$

$$t=0 \quad \frac{3}{2} = \frac{5}{6} \times 0^2 + c$$

$$c = \frac{3}{2}$$

$$v = \frac{5}{6}t^2 + \frac{3}{2} \quad \checkmark$$

$$v = \frac{1}{6} (5t^2 + 9)$$

(ii)

$$x = \frac{5}{18}t^3 + \frac{3}{2}t + c$$

$$t=0, x=0$$

$$\therefore c=0$$

$$x = \frac{5}{18}t^3 + \frac{3}{2}t$$

$$\text{when } t=2$$

$$x = \frac{5}{18}(2)^3 + \frac{3}{2}(2)$$

$$= 5\frac{2}{9}$$

$\therefore$  The displacement is  $5\frac{2}{9} \text{ m}$  (in the positive direction) ✓

(d)

$$\frac{dc}{dt} > 0 \quad \checkmark$$

$$\frac{d^2c}{dt^2} < 0 \quad \checkmark$$

QUESTIONS 6

(a) (i)  $y = 2x^3 - 9x^2 + 12x$   
 $\frac{dy}{dx} = 6x^2 - 18x + 12$  ✓  
 $= 6(x^2 - 3x + 2)$   
 $= 6(x-2)(x-1)$

(ii)  $\frac{dy}{dx} = 0$  for stationary points

$6(x-2)(x-1) = 0$

(2nd derivative method)  $x = 1$  or  $x = 2$  ✓

$\frac{d^2y}{dx^2} = 12x - 18$

$x = 1$

$\frac{d^2y}{dx^2} = 12 \times 1 - 18$

$= -6$

$< 0$

∴ relative maximum turning point at (1, 5)

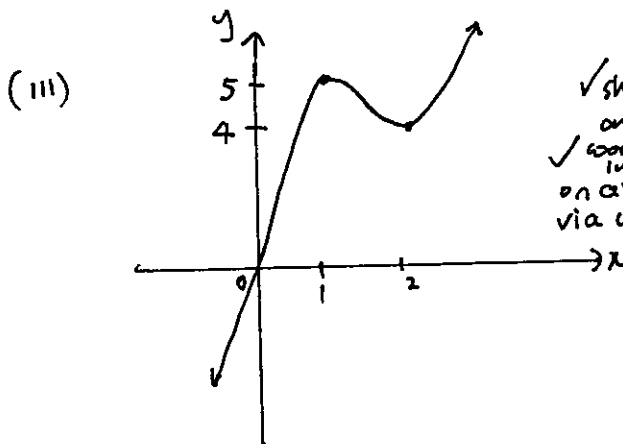
$x = 2$

$\frac{d^2y}{dx^2} = 12 \times 2 - 18$

$= 6$

$> 0$

∴ relative minimum turning point at (2, 4)



✓ shape and coordinates indicated on axes or via wordnotes

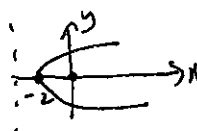
(iv)  $1 < x < 2$  ✓

(b) (i) (-2, 0) ✓

(ii)  $8 = 4a$   
 $a = 2$

Focus (0, 0)

(iii)  $x = -4$  ✓



(c)  $\Delta = 0$

$m^2 - 4 \times 4 \times q = 0$  ✓

$m^2 = 144$

$m = \pm 12$

✓ must have both

or (1st derivative test)

| x     | 0  | 1 | 1.5  | 2 | 3  |
|-------|----|---|------|---|----|
| f(x)  | 12 | 0 | -1.5 | 0 | 12 |
| slope | ✓  | - | ✓    | - | ✓  |

note they need to be careful testing to the right of  $x=1$  and to the left of  $x=2$

∴ relative maximum turning point at (1, 5)  
 relative minimum turning point at (2, 4)

1 mark for correct test and a pair of coordinates (each time)

or

1 mark for 2 correct tests (i.e. mistakes only in y-values)

but

not 1 mark for 2 y values

# QUESTION 7

$$\begin{aligned} \text{(11)} \quad \frac{dp}{dt} &= K P \\ &= \frac{1}{2} \ln\left(\frac{5}{4}\right) \times 61035.156... \checkmark \\ &= 6809.80076... \end{aligned}$$

$$\begin{aligned} \text{or using (11) answer} \\ &= \frac{1}{2} \ln \frac{5}{4} \times 61000 \quad (\checkmark) \\ &= 6805.878... \end{aligned}$$

$$\begin{aligned} \text{or using (11) answer + rounded } K \\ &= 0.1116 \times 61000 (\checkmark) \\ &= 6807.6 \end{aligned}$$

Answer for various "K"  
Answer is correct to nearest hundred so  $\rightarrow$  6800 people/year.

Alternate variables

Students might also do (accept bald answer)  
 $P = 20000e^{Kt}$   
 $\frac{dp}{dt} = 20000Ke^{Kt}$  (accept with a sub for (K))  
 $= \dots$  (answers same as above)

$$\text{(b)} \quad x = t + \frac{16}{t+1}$$

$$\begin{aligned} \text{(i)} \quad t &= 0 \\ x &= 0 + \frac{16}{0+1} \\ x &= 16 \quad \checkmark \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad v &= 1 - 16(t+1)^{-2} \checkmark \\ &= 1 - \frac{16}{(t+1)^2} \end{aligned}$$

$$\begin{aligned} a &= 32(t+1)^{-3} \checkmark \\ &= \frac{32}{(t+1)^3} \end{aligned}$$

$$\text{(iii)} \quad 0 = 1 - \frac{16}{(t+1)^2}$$

$$\begin{aligned} \frac{16}{(t+1)^2} &= 1 \\ (t+1)^2 &= 16 \end{aligned}$$

$$\begin{aligned} t+1 &= \pm 4 \\ t &= 3 \text{ or } -5 \end{aligned}$$

$$t \geq 0 \quad \therefore t = 3 \quad \checkmark$$

(a)

$$\frac{dp}{dt} = KP$$

$$\text{(i)} \quad P = 20000e^{Kt}$$

$$25000 = 20000e^{2K}$$

$$e^{2K} = \frac{25}{20}$$

$$= \frac{5}{4}$$

$$2K = \log_e\left(\frac{5}{4}\right)$$

$$K = \frac{1}{2} \log_e\left(\frac{5}{4}\right)$$

$$= 0.111571...$$

$$= 0.1116 \quad (\text{4 sig. fig})$$

$$\text{(ii)} \quad P = 20000e^{Kt}$$

$$t = 10, \quad K = \frac{1}{2} \log_e\left(\frac{5}{4}\right)$$

$$P = 20000e^{5 \ln \frac{5}{4}}$$

$$= 61035.156...$$

$\therefore$  Population is 61000 (correct to nearest hundred)

Using  $K = 0.1116$  gives  $P = 61052.38...$  and 61100 is population

$$= 7$$

(iv)  $t \rightarrow \infty$

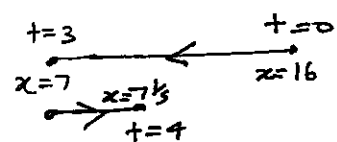
$\therefore$  limiting velocity is  $1 \text{ m s}^{-1}$  ✓ (accept 1)

alternate method  
and mark allocation.

OR

$$= 7\frac{1}{5}$$

$\therefore$  Total distance travelled  $= 9 + \frac{1}{5}$   
 $= 9\frac{1}{5} \text{ m}$  (✓)



}(✓)

# QUESTION 8

$$(a) \frac{dy}{dx} = \frac{2x}{x^2+e}$$

$$y = \int \frac{2x}{x^2+e} dx$$

$$y = \log_e(x^2+e) + C$$

(0, 2)

$$2 = \log_e e + C$$

$$C = 1$$

$$\therefore y = \log_e(x^2+e) + 1$$

$$(b)(i) y = 4 \ln x$$

$$\frac{y}{4} = \log_e x$$

$$x = e^{y/4}$$

$$V = \pi \int_0^4 (e^{y/4})^2 dy$$

$$= \pi \int_0^4 e^{0.5y} dy$$

for 2nd mark  
look for squaring of  
fraction and calculation  
of 4ln e = 4

$$\begin{cases} x=1 \\ y=0 \\ x=e \\ y=4 \ln e \\ =4 \end{cases}$$

$$(ii) V = \pi \left[ \frac{e^{y/2}}{1/2} \right]_0^4$$

$$= \pi(2e^2 - 2 \times 1)$$

$$= 2\pi(e^2 - 1) \text{ units}^3$$

$$(c)(i) \log 3 + \log 9 + \log 27 + \dots$$

$$= \log 3 + 2\log 3 + 3\log 3 + \dots$$

$$T_2 - T_1 = 2\log 3 - \log 3$$

$$= \log 3$$

$$T_3 - T_2 = 3\log 3 - 2\log 3$$

$$= \log 3$$

$$T_3 - T_2 = T_2 - T_1 \therefore \text{A.P.}$$

$$(ii) S_n = \frac{n}{2} [2a + (n-1)d]$$

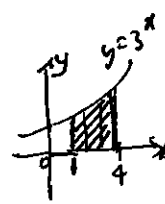
$$S_{10} = \frac{10}{2} [2 \times \log 3 + (10-1) \times \log 3]$$

$$= \frac{10}{2} [2 \log 3 + 9 \log 3]$$

$$= 55 \log 3$$

(d)

|      |                |                |                |                |
|------|----------------|----------------|----------------|----------------|
| x    | 1              | 2              | 3              | 4              |
| A(n) | 3 <sup>1</sup> | 3 <sup>2</sup> | 3 <sup>3</sup> | 3 <sup>4</sup> |



or equivalent

$$\text{Area} = \frac{1}{2} [3^1 + 2 \times 3^2 + 2 \times 3^3 + 3^4]$$

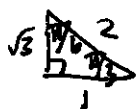
$$= 78 \text{ units}^2$$

QUESTION 9

(a)  $\cos x = -\frac{\sqrt{3}}{2}$

$x = \frac{5\pi}{6}, -\frac{5\pi}{6}$

✓✓



(b)(i)  $f(x) = e^{\tan x}$

$f'(x) = \sec^2 x e^{\tan x}$

✓

(ii)  $\therefore \int 3e^{\tan x} \sec^2 x dx = 3e^{\tan x} + c$

$\int_0^{\pi/4} 3e^{\tan x} \sec^2 x dx = [3e^{\tan x}]_0^{\pi/4}$

✓

$= 3(e^{\tan \pi/4} - e^0)$

$= 3(e - 1)$

✓

(c) (i)  $240^\circ$

$= 240^\circ \times \frac{\pi}{180}$

$= \frac{4\pi}{3}$

$\left. \begin{array}{l} \text{or } \frac{2}{3} \times 2\pi \\ = \frac{4\pi}{3} \end{array} \right\}$

✓

(ii)  $L = r\theta$

$= 12 \times \frac{4\pi}{3}$

$= 16\pi \text{ cm}$

✓

(iii)  $a = \frac{1}{2} r^2 \theta$

$= \frac{1}{2} \times 12^2 \times \frac{4\pi}{3}$

$= 96\pi \text{ cm}^2$

✓

(d) (i)  $a = 1$

$n = 2$

} need both

(d)(ii)  $\text{Area} = \int_{\pi/2}^{3\pi/2} (\sin 2x - 4 \cos x) dx$

$= \left[ -\frac{1}{2} \cos 2x - 4 \sin x \right]_{\pi/2}^{3\pi/2}$

✓

$= \left( -\frac{1}{2} \cos 3\pi - 4 \sin 3\pi \right) - \left( -\frac{1}{2} \cos \pi - 4 \sin \pi \right)$

$= \left( -\frac{1}{2} \times -1 - 4 \times -1 \right) - \left( -\frac{1}{2} \times -1 - 4 \times 0 \right)$

$= \left( \frac{1}{2} + 4 \right) - \left( \frac{1}{2} - 0 \right)$

$= 4\frac{1}{2} - \left( -3\frac{1}{2} \right)$

$= 8 \text{ units}^2$

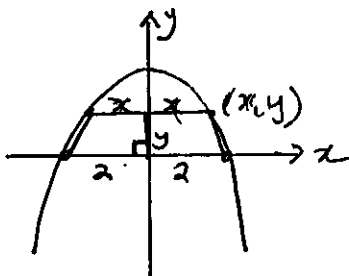
✓

# QUESTION 10

(a)  $\int_{-2}^5 g(y) dy = 7-9$   
 $= -2$  ✓

(b) (i)  $\left. \begin{matrix} R(-2,0) \\ S(2,0) \end{matrix} \right\}$  ✓

(ii)



Area =  $\frac{1}{2} \times y \times (2x+4)$  ✓ as required  
 $= \frac{1}{2} \times (4-x^2) \times (2x+4)$  ✓  
 $= (x+2)(4-x^2)$

as required

(iii)  $A = 4x - x^3 + 8 - 2x^2$   
 $= -x^3 - 2x^2 + 4x + 8$

$\frac{dA}{dx} = -3x^2 - 4x + 4$  ✓

$\frac{dA}{dx} = 0$

$3x^2 + 4x - 4 = 0$

$(3x-2)(x+2) = 0$

$x = \frac{2}{3}$  or  $-2$

$x \neq -2$

$x = \frac{2}{3}$  ✓

$\frac{d^2A}{dx^2} = -6x - 4$

$x = \frac{2}{3}$

$\frac{d^2A}{dx^2} = -6 \times \frac{2}{3} - 4$

$= -8$

$< 0$  ✓  $\therefore$  maximum value of A when  $x = \frac{2}{3}$

| x               | $\frac{1}{3}$ | $\frac{2}{3}$ | 1  |
|-----------------|---------------|---------------|----|
| $\frac{dA}{dx}$ | $\frac{2}{3}$ | 0             | -3 |
| Sign            | +             | -             | -  |

(c)  $B_1 = 50\,000 \times 1.05 - 1500$  ✓

$B_2 = (50\,000 \times 1.05 - 1500) \times 1.05$   
 $- 1500 \times 1.03$  ✓

$= 50\,000 \times 1.05^2 - 1500(1.05 + 1.03)$

$B_3 = [50\,000 \times 1.05^2 - 1500(1.05 + 1.03)] \times 1.05$   
 $- 1500(1.03)^2$

$= 50\,000 \times 1.05^3 - 1500(1.05^2 + 1.03 \times 1.05 + 1.03^2)$

Note  $B_2$  and  $B_3$   
 Marks awarded for indicating the method on previous "B"

and not just the answer  
 (especially  $B_3$  as it is given in the question!)

(ii) Following this pattern

$B_n = 50\,000 \times 1.05^n$   
 $- 1500(1.05^{n-1} + 1.03 \times 1.05^{n-2} + 1.03^2 \times 1.05^{n-3} + \dots + 1.03^{n-1})$

G.P.  $a = 1.05^{n-1}$

$r = \frac{1.03}{1.05}$

'n' terms

Using  $S_n = \frac{a(r^n - 1)}{r - 1}$

naive

showing how!!

$B_n = 50\,000 \times 1.05^n - 1500 \left[ \frac{1.05^{n-1} \left( \left( \frac{1.03}{1.05} \right)^n - 1 \right)}{\frac{1.03}{1.05} - 1} \right]$

$= 50\,000 \times 1.05^n - 1500 \left[ \frac{1.05^{n-1} \left( \left( \frac{1.03}{1.05} \right)^n - 1 \right)}{\frac{-0.02}{1.05}} \right]$

$= 50\,000 \times 1.05^n - \frac{1500}{0.02} \left[ 1.05^n \left( 1 - \left( \frac{1.03}{1.05} \right)^n \right) \right]$

$= 50\,000 \times 1.05^n - 75\,000 \times 1.05^n \left( 1 - \left( \frac{1.03}{1.05} \right)^n \right)$

given in question AS REQUIRED!